Introduction to Neural Networks

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Abstract

This document provides an overview of neural networks, including their structure, mathematical representation, training process and common notations used in the field.

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1 Overview

Neural networks are computational models inspired by the human brain. They consist of interconnected nodes, or **neurons**, arranged in layers. The basic components of a neural network include:

- Input Layer
- Hidden Layers
- Output Layer

2 Mathematical Representation

Mathematically, a neuron processes input x_i with associated weights w_i and a bias b:

$$y = f\left(\sum_{i=1}^{n} w_i x_i + b\right) \tag{1}$$

where f is the activation function. Common choices for f include:

2.1 Activation Functions

1. Sigmoid: $f(x) = \frac{1}{1+e^{-x}}$
2. ReLU: $f(x) = \max(0, x)$
3. Tanh: $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

3 Training Process

The learning process involves updating the weights using an optimization algorithm like *gradient descent*. Given a loss function L, the weight update follows:

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w} \tag{2}$$

$$b_{new} = b_{old} - \eta \frac{\partial L}{\partial b} \tag{3}$$

Equation (1) describes the neuron function, and the weight update rule is shown in (2).

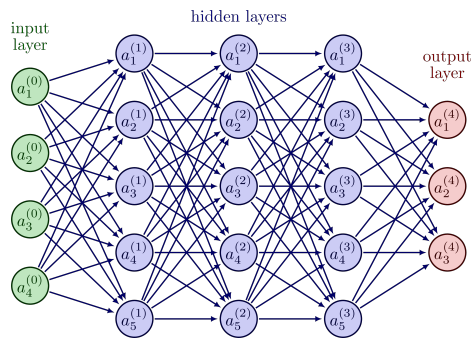
4 Greek Letters in Neural Networks

Neural networks often use Greek letters for notation:

- α (learning rate)
- θ (model parameters)
- λ (regularization parameter)

5 Illustration

An example of a neural network structure is shown below:



6 Neural Network Parameters

The following table summarizes key parameters of a neural network:

Parameter	Description
w	Weights of connections
b	Bias term
α	Learning rate
L	Loss function

Table 1: Neural Network Parameters

7 Conclusion

Neural networks are a powerful tool for pattern recognition and machine learning. Understanding their structure and training mechanisms is crucial for implementing AI systems.

References

[1] Y. LeCun, Y. Bengio, and G. Hinton. "Deep learning." *Nature*, vol. 521, no. 7553, 2015, pp. 436-444.

[2] I. Goodfellow, Y. Bengio, and A. Courville. *Deep Learning*. MIT Press, 2016.

[3] D. P. Kingma and J. Ba. "Adam: A method for stochastic optimization." *International Conference on Learning Representations (ICLR)*, 2015.

[4] X. Glorot and Y. Bengio. "Understanding the difficulty of training deep feedforward neural networks." *International Conference on Artificial Intelligence and Statistics (AISTATS)*, Sardinia, Italy, 2010.

[5] A. Krizhevsky, I. Sutskever, and G. Hinton. "ImageNet classification with deep convolutional neural networks." *Advances in Neural Information Processing Systems (NeurIPS)*, 2012.